Introduction to Microcontrollers

Facilitator Guide

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Week 2: Getting Started with Mixly

Objective:

By the end of this lesson, students will:

Understand the basic features of Mixly, a block-based programming environment for Arduino.

Be able to set up Mixly, connect it to an Arduino board, and create simple programs.

Upload and run a simple program using Mixly.

Lesson Outline:

Introduction to Mixly (10 minutes)

What is Mixly?

Why use Mixly instead of the Arduino IDE?

Key features of Mixly.

Installing Mixly and Connecting to Arduino (15 minutes)

Step-by-step guide to downloading and installing Mixly.

Setting up Mixly with an Arduino board.

Overview of Mixly's interface.

Basic Programming in Mixly (20 minutes)

Understanding basic programming blocks.

Hands-on activity: Creating a simple LED blink program in Mixly.

Uploading and Running Programs (10 minutes)

Uploading the LED blink program to the Arduino.

Debugging and troubleshooting common issues.

Q&A and Review (5 minutes)

Detailed Lesson Plan:

1. Introduction to Mixly (10 minutes)

What is Mixly?

Mixly is a graphical programming environment based on block programming, designed for use with Arduino boards.

It simplifies coding by allowing users to drag and drop code blocks to form programs instead of writing lines of code. This makes it ideal for beginners and those unfamiliar with text-based programming.

Why use Mixly?

Visual learning: Users can understand programming concepts without learning syntax.

Easy prototyping: Quick to assemble and modify code for Arduino projects.

Real-time interaction: It provides an instant understanding of how blocks work in Arduino programming.

Key Features of Mixly:

Drag-and-drop interface: Simplifies coding with no need for text-based syntax.

Real-time coding feedback: Visual blocks represent actual code in real-time.

Easy connection to Arduino: Designed for fast prototyping with Arduino boards.

2. Installing Mixly and Connecting to Arduino (15 minutes)

Step-by-step Guide to Downloading and Installing Mixly:

Go to the Mixly download page or search for it online.

Choose the version appropriate for your operating system (Windows, macOS, Linux).

Download the ZIP file and extract the contents to your computer.

Open the Mixly application (no installation required).

Setting up Mixly with an Arduino Board:

Connect your Arduino (e.g., Uno, Nano, Mega) to your computer using a USB cable.

Open Mixly and go to the Tools menu.

Select the appropriate COM port and board type from the Tools menu.

Overview of Mixly's Interface:

Programming workspace: Main area where blocks are placed to create programs.

Block categories: Sidebar on the left with various blocks like Control, Logic, Sensors, and more.

Upload button: Located at the top-right, it uploads the program to the Arduino board.

3. Basic Programming in Mixly (20 minutes)

Understanding Basic Programming Blocks:

Control Blocks: Used to control the flow of a program (e.g., loops, conditions).

Logic Blocks: Handle logical operations such as true/false and comparisons.

Pin Blocks: Control the input and output pins on the Arduino board.

Hands-on Activity: Creating a Simple LED Blink Program in Mixly:

Step 1: Open Mixly.

Drag the control block labeled "forever" into the workspace.

Step 2: Add a Pin Block.

From the "Pin" section, drag the "digitalWrite" block inside the forever loop.

Set the pin number to 13 (the built-in LED on Arduino boards).

Step 3: Control LED State.

Inside the "digitalWrite" block, select HIGH to turn the LED on.

Add a delay block from the "Control" section and set it to 1000 ms.

Step 4: Add a Turn-off Sequence.

Repeat the above steps but set the digitalWrite block to LOW to turn the LED off.

Step 5: Test the Program.

Ensure that the program will alternate between turning the LED on and off every second.

4. Uploading and Running Programs (10 minutes)

Uploading the LED Blink Program to the Arduino:

Ensure your Arduino is connected to the computer.

Click the Upload button in Mixly to send the program to your Arduino board.

Check the output window for confirmation that the upload was successful.

Debugging and Troubleshooting:

If the upload fails:

Ensure the correct COM port and board type are selected.

Check the USB connection.

Re-upload the program after fixing the issues.

5. Q&A and Review (5 minutes)

Recap the key steps for creating and uploading a program in Mixly.

Answer any questions students may have about using Mixly or the Arduino setup.

Discuss possible next steps for more advanced projects.

Conclusion:

This one-hour lesson introduces students to Mixly, a block-based programming tool, and teaches them how to set up their Arduino and upload simple programs. Through hands-on activities, students will gain the confidence to start working on their own Arduino-based projects.

For more about mixly you can read this link:
https://wiki.keyestudio.com/Getting_Started_with_Mixly